

Compositional monitoring of energy-mix drift on the simplex

Which carrier did the structural work?

Standard compositional data analysis — what happens when we read the simplex (shares that sum to one) forward in time.

Follow along on the repository — the slide deck, manuscript, and live projector are all open.

H^s runs any compositional dataset the CoDa community can describe — every view here is reproducible on your data.

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github.com / PeterHiggins19 / higgins-decomposition

Community folder: CODA-Association/ · Hand-out in UN-6 locales (EN · FR · ES · RU · ZH · AR)

The instrument reads. The expert decides. The hashes carry the receipts. The vocabulary holds the line.

The size view hides the work

A carrier can be small in share yet do most of the structural work — its share of the squared compositional motion.

Germany electricity, 25 years

the size view — the standard share / stacked-area chart

Coal and lignite recede.

Nuclear phases out.

Gas holds.

Solar and wind grow steadily.

Wind passes coal in the late 2010s.

What the size view misses

Germany Solar, 2005 → 2006

starting share	0.21 %
structural Power Share	71.1 %
Activation Coefficient	$\approx 333 \times$

*Solar acted at $333 \times$ its size,
four years before the share view
calls solar visible.*

Deceptive drift: the size view hides which carrier did the structural work — here a 0.21 % carrier does 71 % of it.

Mathematics: standard compositional data analysis. Application: monitoring frame may be new.

Reading time on the simplex

Standard CoDa geometry, read forward in time — five named readings, each defined below.

INPUT · EMBER electricity-generation data · carriers × N years per country

Closure → **Centred log-ratio (CLR)** → **Helmert-orthonormal ILR** → **Aitchison-distance trajectory**

1. Composition

each carrier's
share, year by year

2. Helmsman

the carrier with the
largest CLR move
at a step

3. Helmsman trajectory

when the largest-
motion carrier
changes

4. Power Share

each carrier's share
of the squared
CLR motion

5. Activation Coefficient

Power Share ÷
starting share

Combined into one observable stack — a falsifiable claim

**the path the
composition took**

(Composition + trajectory)

**why the path
took that shape**

(Activation Coefficient)

**where the hidden
drivers are**

(Power Share concentration)

Five readings · one observable stack · reproducible from the raw CSV

FIVE READINGS

1. Composition

each carrier's share, year by year

2. Helmsman

largest CLR move at a step

3. Helmsman trajectory

when the largest-motion carrier changes

4. Power Share

share of squared CLR motion

5. Activation Coefficient

Power Share ÷ starting share

one observable stack — falsifiable

The Activation Coefficient

How much structural work a carrier does, relative to how much of the mix it is.

$$\alpha_i(t) = \text{Power Share}_i(t) \div \text{starting share}_i(t)$$

$\alpha \approx 1$ carrier does work proportional to its size — ordinary

$\alpha \gg 1$ carrier acts far above its size — hidden driver

$\alpha < 1$ carrier carries less work than its size suggests — coasting

Worked example — Germany Solar 2005 → 2006

starting share	0.21 %	small
Power Share	71.1 %	most of the work
α	$\approx 333 \times$	hidden driver

Analogy — yeast is 2 % of a loaf by mass and does 100 % of the rising: the same shape.

The Energiewende's structural beginning — four years before solar appears in the share view.

Three archetypes — one instrument, three regimes

The same protocol on three transitions that look fundamentally different.

Germany

deliberate transition

continuous arc

Energiewende

2000 → 2025

solar + wind absorb

structural work

before size dominates.

Japan

external shock

loop and reorganise

Fukushima 2011

displaces nuclear,
causes 2011–2013

multi-year

reorganisation.

United Kingdom

regime change

jump and return

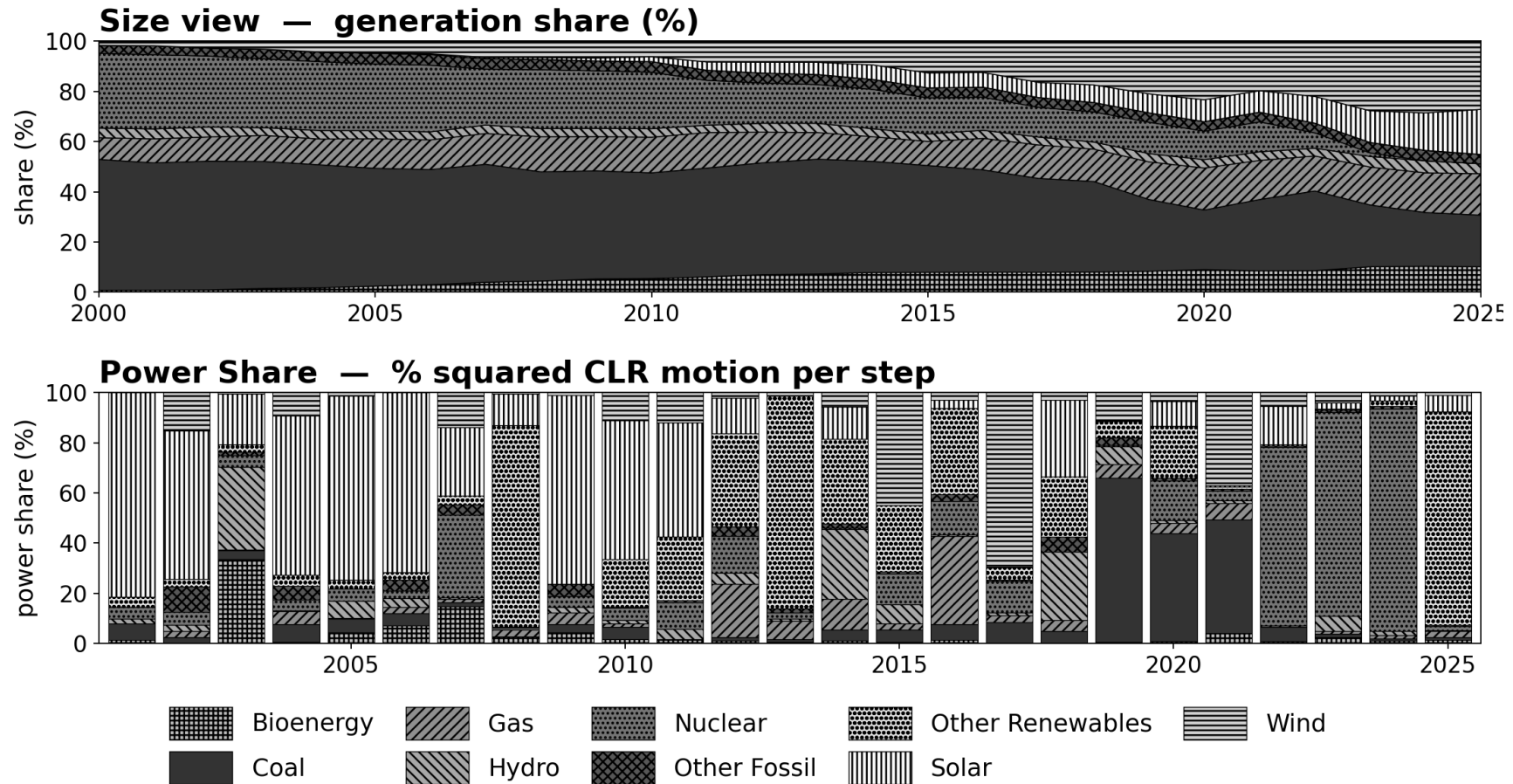
Coal exit 2012–2020
from > 30 % to < 2 %.

Wind, solar, others
absorb displaced
structural work.

Three different transition regimes. One operational protocol reads them all.

Germany — share and structural work

The Energiewende read as a single smooth arc on the simplex.



Solar 2005-2006: 0.21 % share • 71.1 % structural work • $\alpha \approx 333 \times$

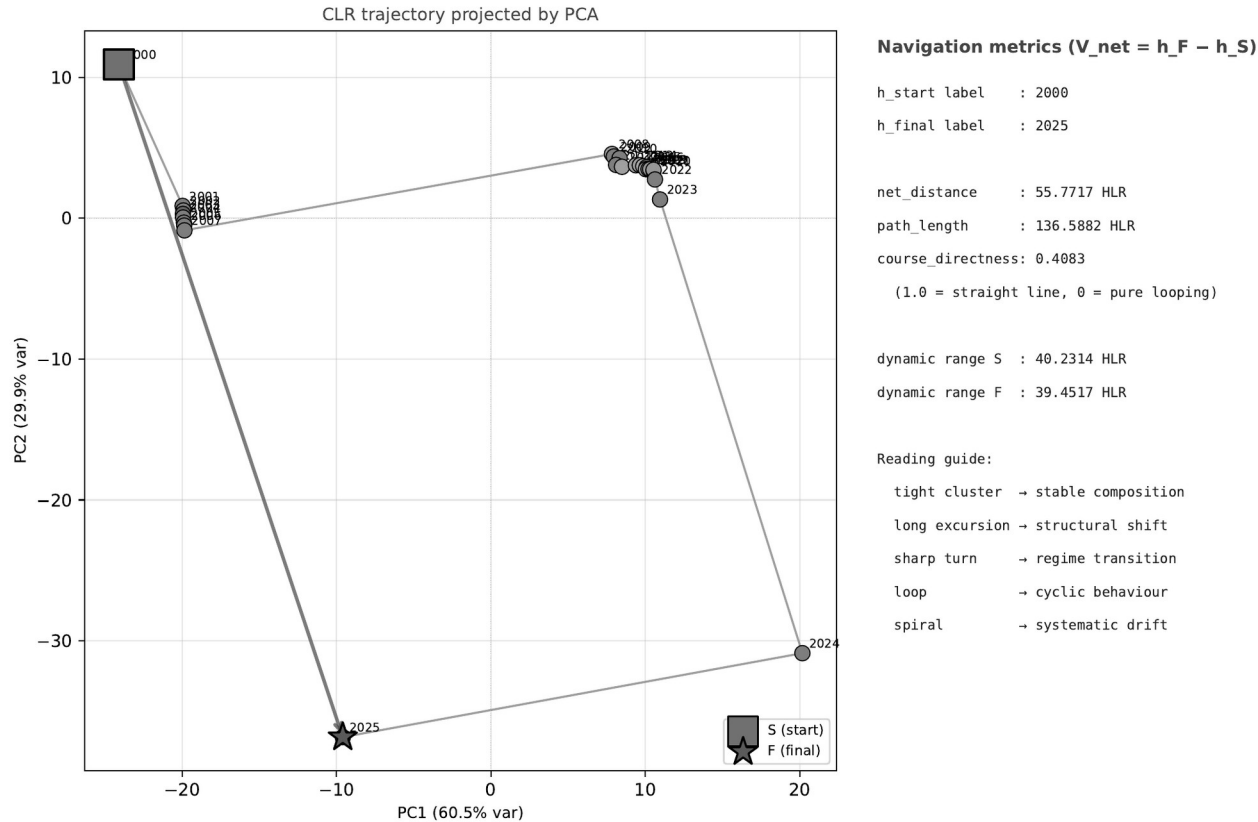
Tiny share, dominant work — the deceptive-drift signature, four years before the share view shows anything.

Germany — the trajectory

The Helmsman trajectory: where the composition went, year by year.

Compositional system geometry and dynamics for, Germany — EMBER electricity generation 2000-2025

Plate 16 | System Course Plot — PCA 2D path with S→F net vector
Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



engine: cnt 2.0.4 (c5111ef761bb...) | data: 2c9471359a4e... | source: 8d52a1c20cd3...
2026-05-06 | run: v1.1.x-final | page 16/28 | STAGE 2 (Order 2) | system course plot (PCA)

DEU

continuous arc

deliberate transition

directness 0.41

no loops

no flips

renewable

vertex

Germany — orthogonal projections (XY / XZ / YZ)

Germany alone gets the complete plate set • the Section view: CLR plan, XZ bearings, and per-carrier CLR for 2013.

SECTION PLATE t=13 | Year 2013 | D=9 N=26 pairs=36

HCI Stage 1 | FIXED SCALE
Unit: HLR (Higgins Log-Ratio)
Y-x=Y/T-h=CLR-section

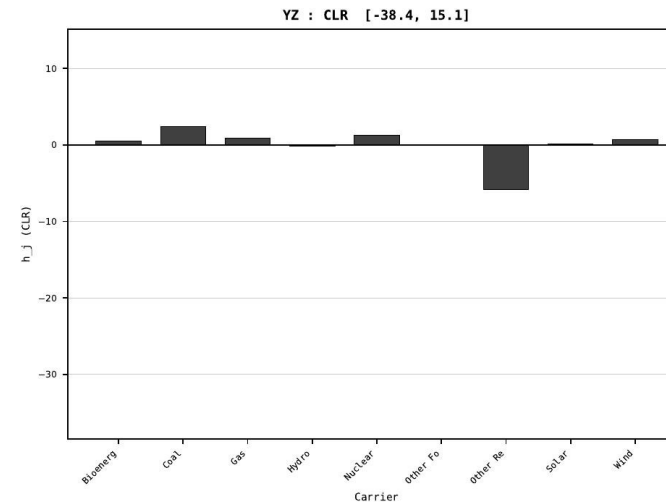
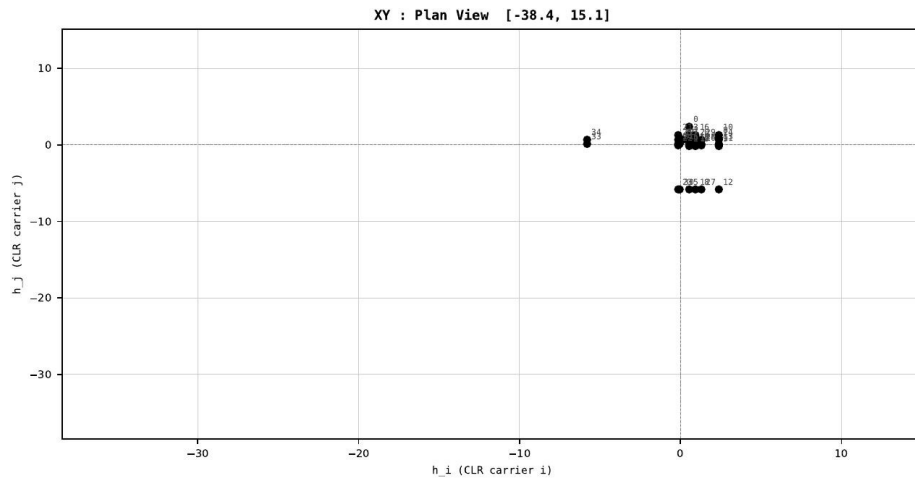
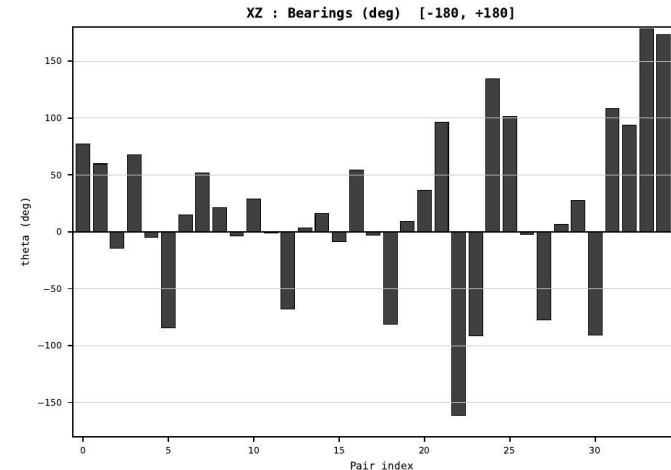
Dataset : ember_DEU_Germany_generation_TWh.csv
Year : 2013
t : 13 / 25
D : 9 carriers
Pairs : 36 channels

Hs = 0.235275
Ring = Hs-2
E_metric= 42.7299
kappa_HS= 3602.50
omega = 2.4484 deg
d_A = 0.925102
Helm = Other Renewables
Helm d = +0.852077
DR = 8.189 HLR
DR ratio= 3602.5x

XZ [-180,180]
CLR [-38.4,15.1]
Hs [0, 1]

PAIR INDEX:

0=Bioenergy-Coal	18=Gas-Other Renewables
1=Bioenergy-Gas	19=Gas-Solar
2=Bioenergy-Hydro	20=Gas-Wind
3=Bioenergy-Nuclear	21=Hydro-Nuclear
4=Bioenergy-Other Fossil	22=Hydro-Other Fossil
5=Bioenergy-Other Renewables	23=Hydro-Other Renewables
6=Bioenergy-Solar	24=Hydro-Solar
7=Bioenergy-Wind	25=Hydro-Wind
8=Coal-Gas	26=Nuclear-Other Fossil
9=Coal-Hydro	27=Nuclear-Other Renewables
10=Coal-Nuclear	28=Nuclear-Solar
11=Coal-Other Fossil	29=Nuclear-Wind
12=Coal-Other Renewables	30=Other Fossil-Other Renewables
13=Coal-Solar	31=Other Fossil-Solar
14=Coal-Wind	32=Other Fossil-Wind
15=Gas-Hydro	33=Other Renewables-Solar
16=Gas-Nuclear	34=Other Renewables-Wind
17=Gas-Other Fossil	35=Solar-Wind



DEU

Section view

XY plan

XZ bearings

YZ per-carrier

H^s 0.235

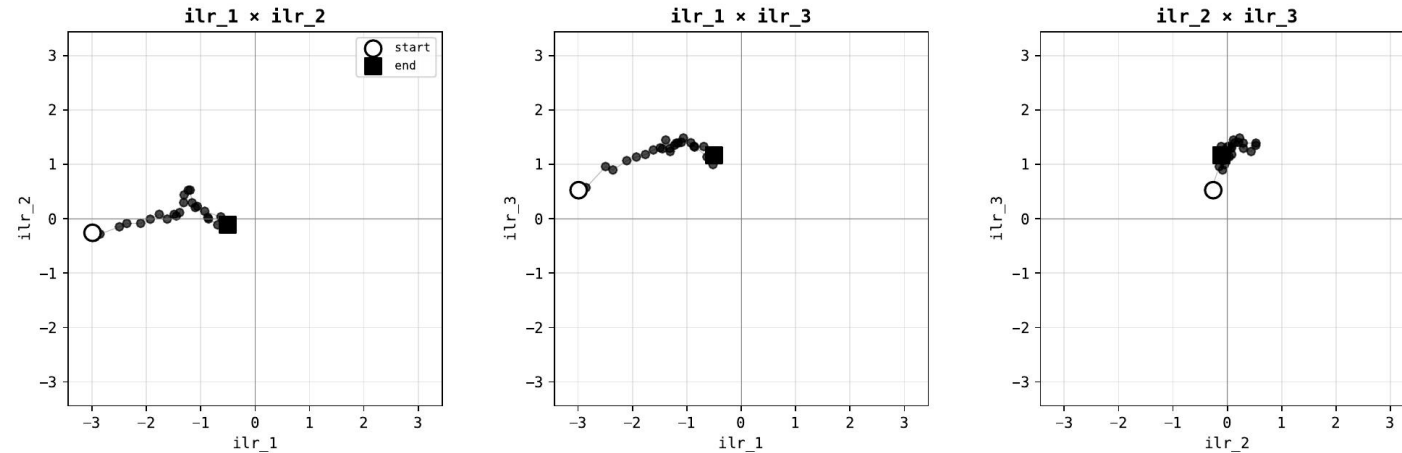
κ 3603

Germany — ILR-Helmert triplet

Germany's complete set • three orthonormal ILR projections of the 2000–2025 trajectory ($ilr_1 \times ilr_2 \cdot ilr_1 \times ilr_3 \cdot ilr_2 \times ilr_3$).

ILR-Helmert Orthogonal Triplet — DEU Germany • Stage 1 Order-1 plate

Three orthogonal scatter projections of CLR onto the first three Helmert ILR axes • N=26 years • D=9 carriers



Helmert basis loadings (which carriers each ILR axis contrasts):

ilr_1: (Bioenergy) vs Coal
ilr_2: (Bioenergy + Coal) vs Gas
ilr_3: (Bioenergy + Coal + Gas) vs Hydro

Trajectory:
start year: 2000
end year: 2025
N readings: 26

Display:
fixed scale ± 3.44 ILR
Order 1 (first principles)
per Output Doctrine v1.0

Symbology:
○ start • ■ end
• intermediate years

DEU

ILR-Helmert Triplet

orthonormal

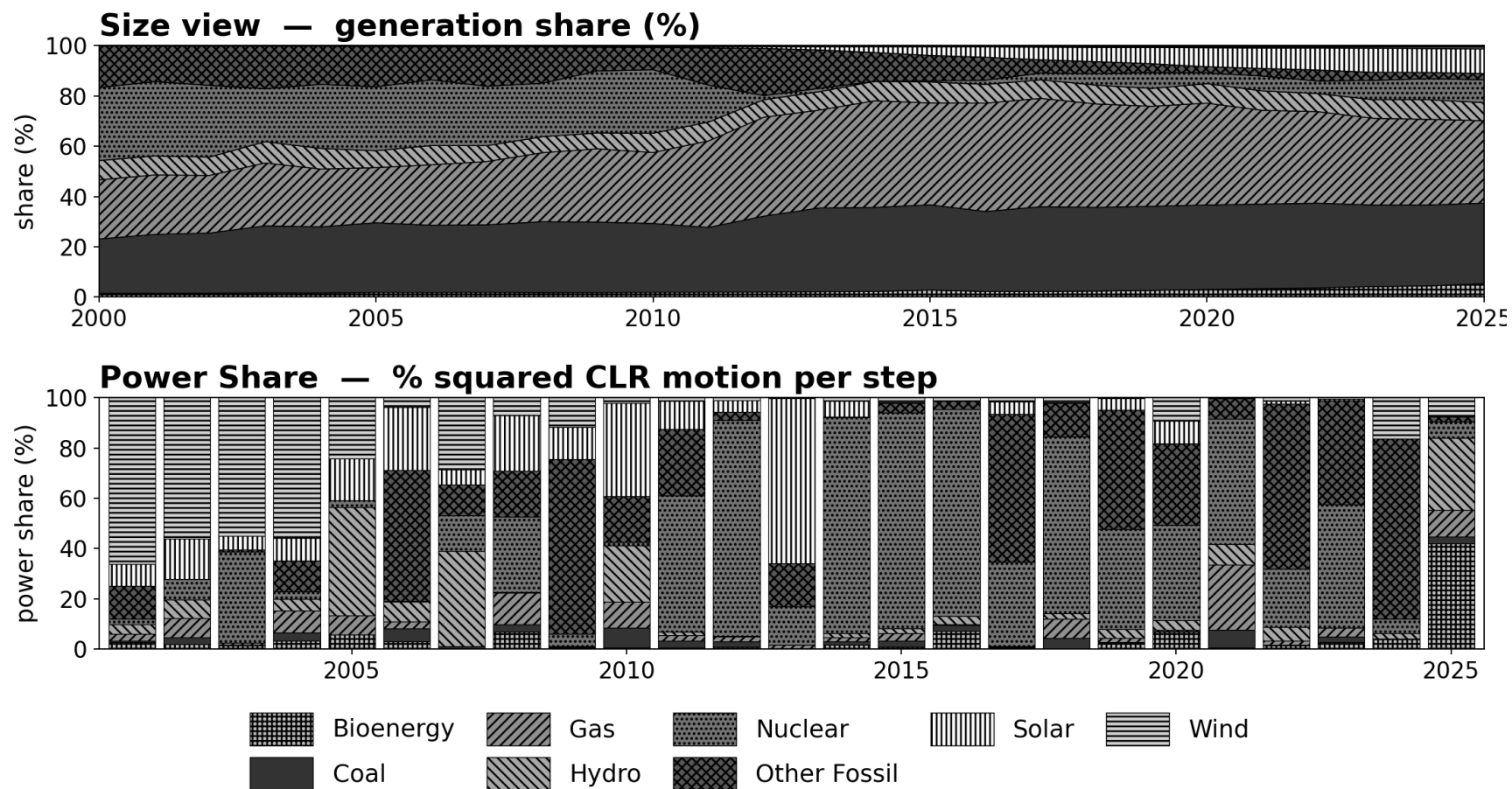
$ilr_1 \times ilr_2$

$ilr_1 \times ilr_3$

$ilr_2 \times ilr_3$

Japan — share and structural work

Fukushima 2011: an external shock displaces nuclear; a multi-year reorganisation follows.



Aitchison distance 2011 → 2012 $\approx 3 \times$ baseline step • 17 Helmsman flips on the corpus

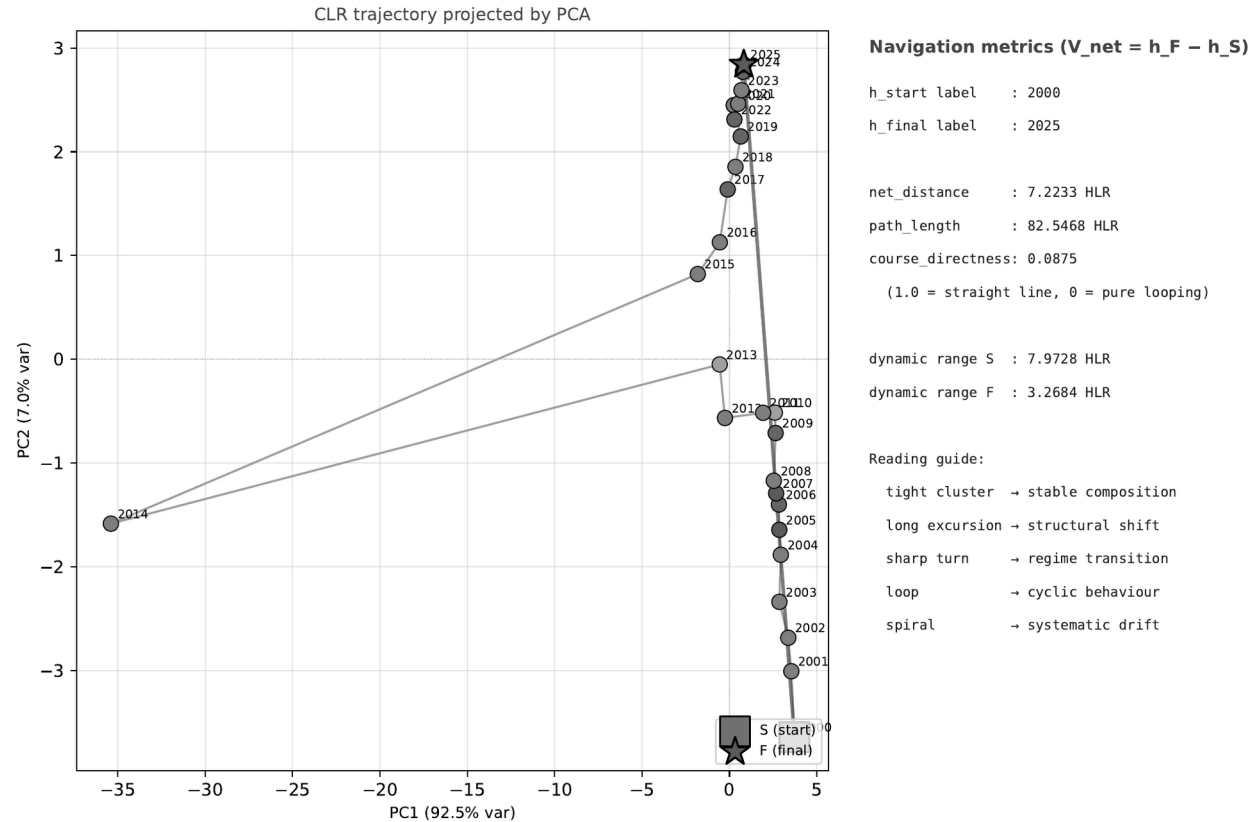
The instrument detects both the shock and the multi-year reorganisation that followed.

Japan — the trajectory

The Helmsman trajectory: looping reorganisation, not a single step.

Compositional system geometry and dynamics for, Japan — EMBER electricity generation 2000-2025

Plate 16 | System Course Plot — PCA 2D path with S→F net vector
Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



engine: cnt 2.0.4 (c5111ef761bb...) | data: f61dcf0026fb... | source: e7643dca648e...
2026-05-06 | run: v1.1.x-final | page 16/28 | STAGE 2 (Order 2) | system course plot (PCA)

JPN

loop and reorganise

external shock

directness 0.09

revisits

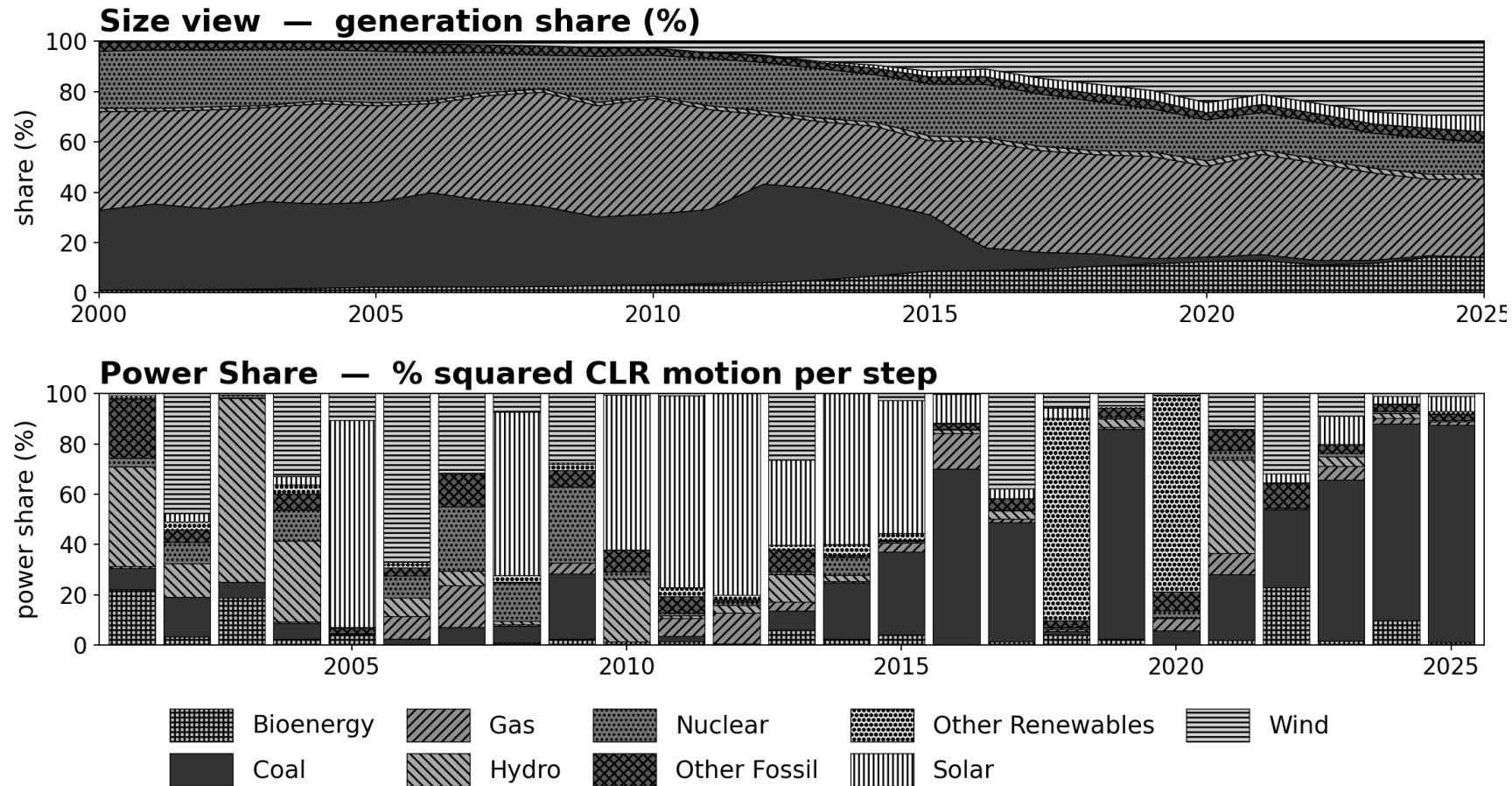
reroutes

search,

not a plan

United Kingdom — share and structural work

Coal exit as policy-driven regime change, absorbed across several carriers.



Coal: > 30 % → < 2 %. Wind, solar, and other renewables absorb the displaced work.

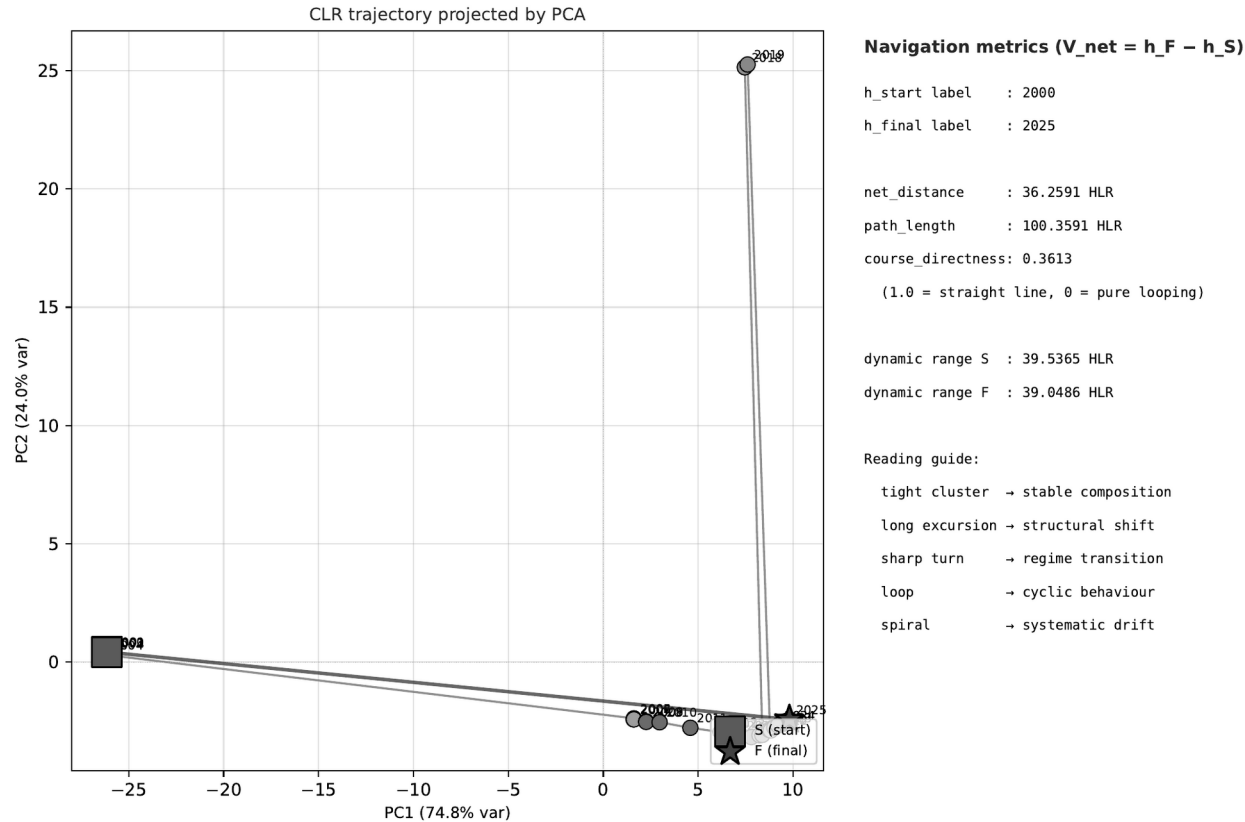
Displaced work redistributes across several carriers — not one substitute, but several.

United Kingdom — the trajectory

The Helmsman trajectory: a jump from the coal vertex, then a return toward stability.

Compositional system geometry and dynamics for, United Kingdom — EMBER electricity generation 2000-2025

Plate 16 | System Course Plot — PCA 2D path with S→F net vector
Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



engine: cnt 2.0.4 (c5111ef761bb...) | data: 56ced72dc4e2... | source: b1df2a0ae6f2...
2026-05-06 | run: v1.1.x-final | page 16/28 | STAGE 2 (Order 2) | system course plot (PCA)

GBR

jump and return

regime change

directness 0.36

sharp departure

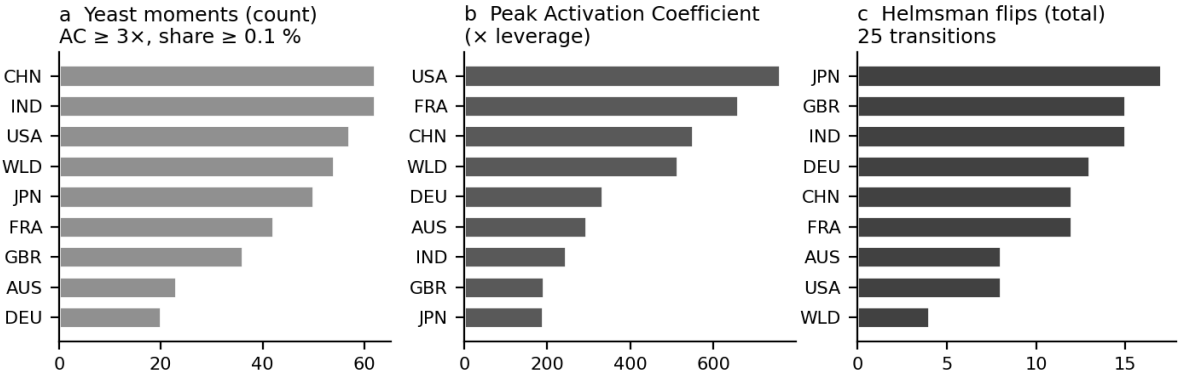
re-stabilisation

one decisive

displacement

Across the corpus — deceptive drift in 5 of 9

The same protocol applied to all nine EMBER countries.



Japan tops helmsman flips (17 — the shock). China & India top yeast count (62 each — small carriers absorbed).
USA, France, World aggregate top peak Activation Coefficient — solar 2010-2015 at 500-760 ×.

d Top ten activation moments across the 9-country corpus						
#	country	transition	carrier	Activation	Power Share	Size at start
1.	USA	2012-2013	Solar	760 ×	81.7 %	0.107 %
2.	FRA	2010-2011	Solar	659 ×	72.6 %	0.110 %
3.	FRA	2004-2005	Wind	634 ×	66.0 %	0.104 %
4.	CHN	2013-2014	Solar	549 ×	84.7 %	0.154 %
5.	WLD	2010-2011	Solar	513 ×	77.7 %	0.151 %
6.	FRA	2005-2006	Wind	495 ×	83.5 %	0.169 %
7.	USA	2013-2014	Solar	395 ×	87.9 %	0.223 %
8.	CHN	2004-2005	Bioenergy	346 ×	39.7 %	0.115 %
9.	CHN	2003-2004	Bioenergy	342 ×	45.5 %	0.133 %
10.	FRA	2024-2025	Coal	342 ×	62.1 %	0.182 %

5 of 9
deceptive drift

PRESENT

AUS · CHN
GBR · IND · JPN

ABSENT

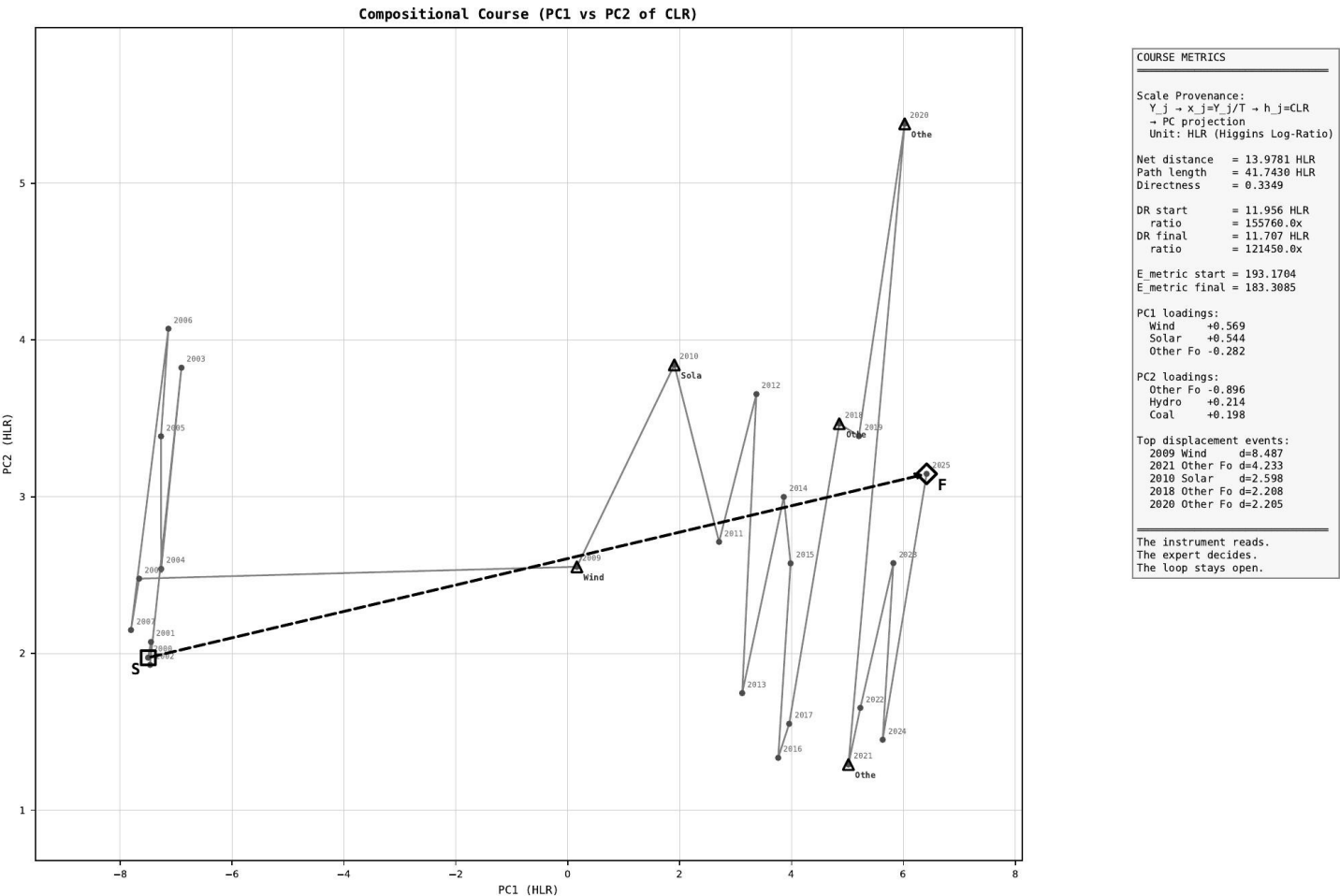
DEU (annual)
FRA · USA · WLD

Deceptive drift: the size view hides which carrier did the structural work — a small carrier carries a large share of the squared compositional motion (Activation Coefficient >> 1).
Discrimination — flagging some systems and not others — is itself evidence the detector reads real structure.

Australia — the trajectory

The rest of the world · deceptive drift present. Australia: the full 2000–2025 trajectory.

SYSTEM COURSE PLOT | ember_AUS_Australia_generation_TWh.csv | D=9 N=26 | Years 2000–2025



AUS

deceptive drift

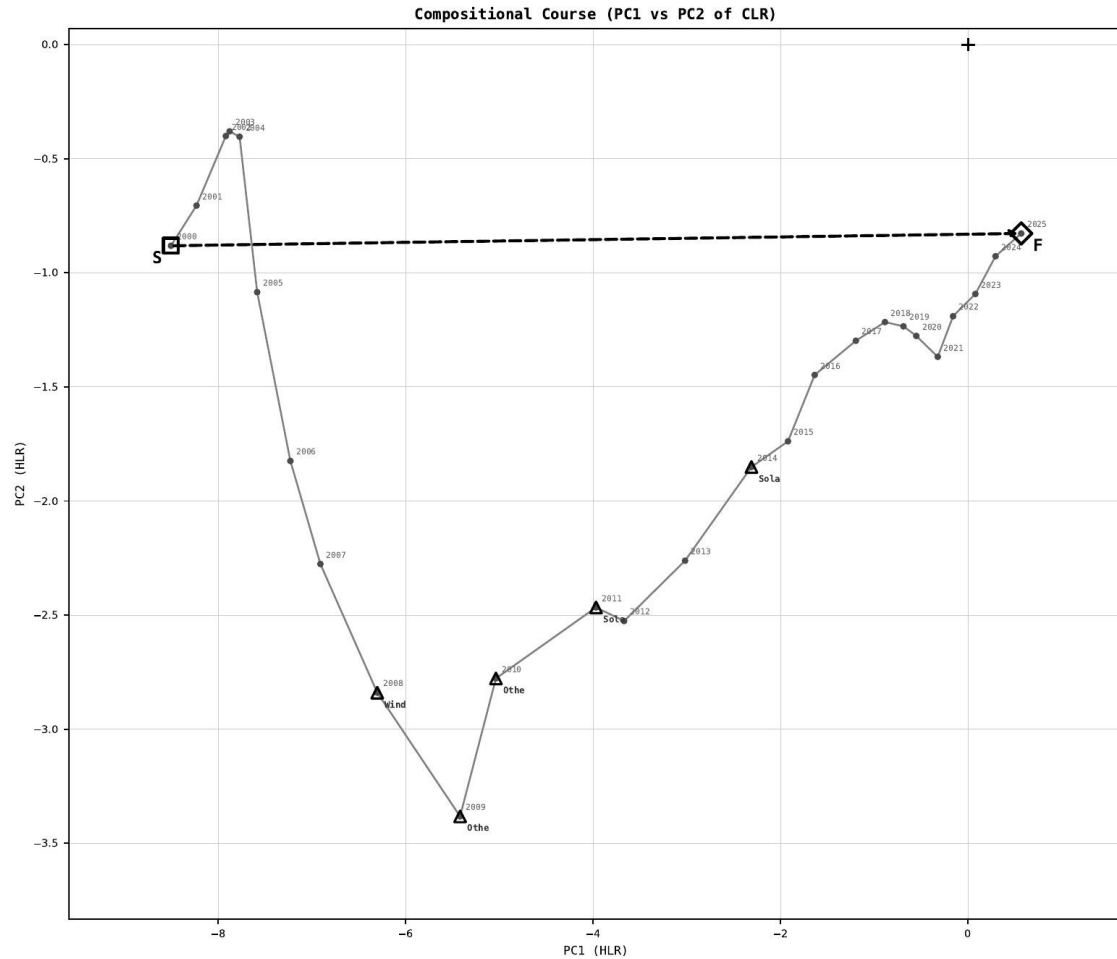
PRESENT

*the rest of the
world (1 of 3)*

China — the trajectory

China: the full trajectory — coal-era growth, then a turn.

SYSTEM COURSE PLOT | ember_CHN_China_generation_TWh.csv | D=8 N=26 | Years 2000–2025



COURSE METRICS	
Scale Provenance: Y_j → x_j=Y_j/T → h_j=CLR → PC projection Unit: HLR (Higgins Log-Ratio)	
Net distance	= 9.0822 HLR
Path length	= 13.8225 HLR
Directness	= 0.6571
DR start	= 10.878 HLR
ratio	= 53013.0x
DR final	= 4.162 HLR
ratio	= 64.2x
E_metric start	= 82.7022
E_metric final	= 11.6993
PC1 loadings:	
Solar	+0.735
Other Fo	-0.396
Wind	+0.347
PC2 loadings:	
Other Fo	+0.557
Solar	+0.502
Wind	-0.472
Top displacement events:	
2011 Solar	d=1.162
2009 Other Fo	d=1.043
2010 Other Fo	d=1.021
2008 Wind	d=0.975
2014 Solar	d=0.855
The instrument reads. The expert decides. The loop stays open.	

CHN

deceptive drift

PRESENT

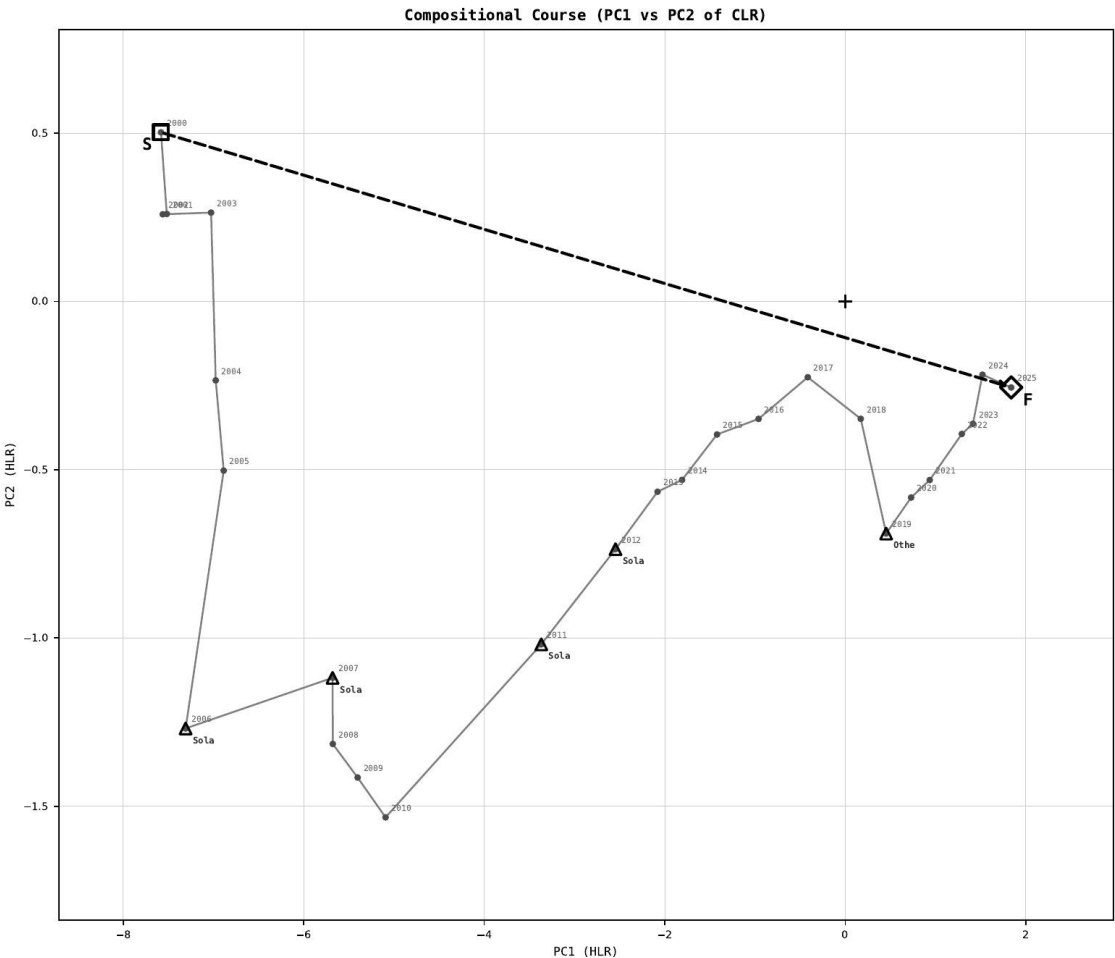
coal-era growth,

then a turn

India — the trajectory

India: the full trajectory — the third case where deceptive drift is present.

SYSTEM COURSE PLOT | ember_IND_India_generation_TWh.csv | D=8 N=26 | Years 2000–2025



COURSE METRICS	
Scale Provenance: Y _j → x _j =Y _j /T → h _j =CLR → PC projection Unit: HLR (Higgins Log-Ratio)	
Net distance	= 9.4603 HLR
Path length	= 14.6188 HLR
Directness	= 0.6471
DR start	= 10.572 HLR
ratio	= 39023.0x
DR final	= 5.085 HLR
ratio	= 359.5x
E _{metric} start	= 75.8817
E _{metric} final	= 20.7772
PC1 loadings:	
Solar	+0.841
Other Fo	-0.425
Gas	-0.241
PC2 loadings:	
Wind	-0.574
Bloenerg	-0.487
Other Fo	+0.458
Top displacement events:	
2011 Solar	d=1.820
2007 Solar	d=1.692
2006 Solar	d=0.943
2012 Solar	d=0.912
2019 Other Fo	d=0.756
The instrument reads. The expert decides. The loop stays open.	

IND

deceptive drift

PRESENT

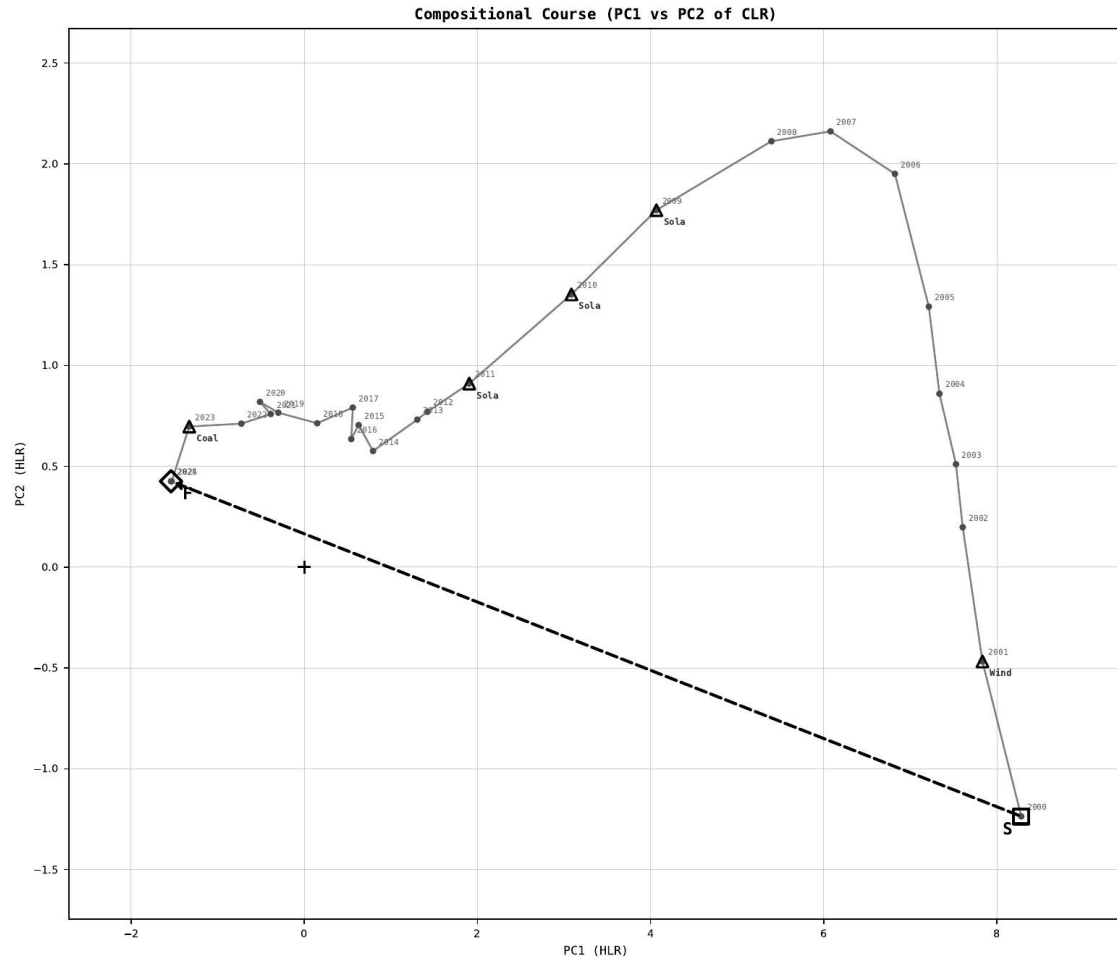
the third case

of present

France — the trajectory

Deceptive drift absent. France: the full trajectory — nuclear-stable.

SYSTEM COURSE PLOT | ember_FRA_France_generation_TWh.csv | D=9 N=26 | Years 2000–2025



COURSE METRICS	
Scale Provenance: Y_j → x_j=Y_j/T → h_j=CLR → PC projection Unit: HLR (Higgins Log-Ratio)	
Net distance	= 10.0035 HLR
Path length	= 17.9314 HLR
Directness	= 0.5579
DR start	= 10.634 HLR
ratio	= 41516.0x
DR final	= 6.534 HLR
ratio	= 687.8x
E_metric start	= 94.4242
E_metric final	= 29.9972
PC1 loadings:	
Solar	-0.768
Coal	+0.387
Wind	-0.385
PC2 loadings:	
Wind	+0.841
Solar	-0.448
Hydro	-0.160
Top displacement events:	
2009 Solar	d=1.437
2011 Solar	d=1.337
2010 Solar	d=1.129
2023 Coal	d=1.095
2001 Wind	d=0.998
The instrument reads. The expert decides. The loop stays open.	

FRA

deceptive drift

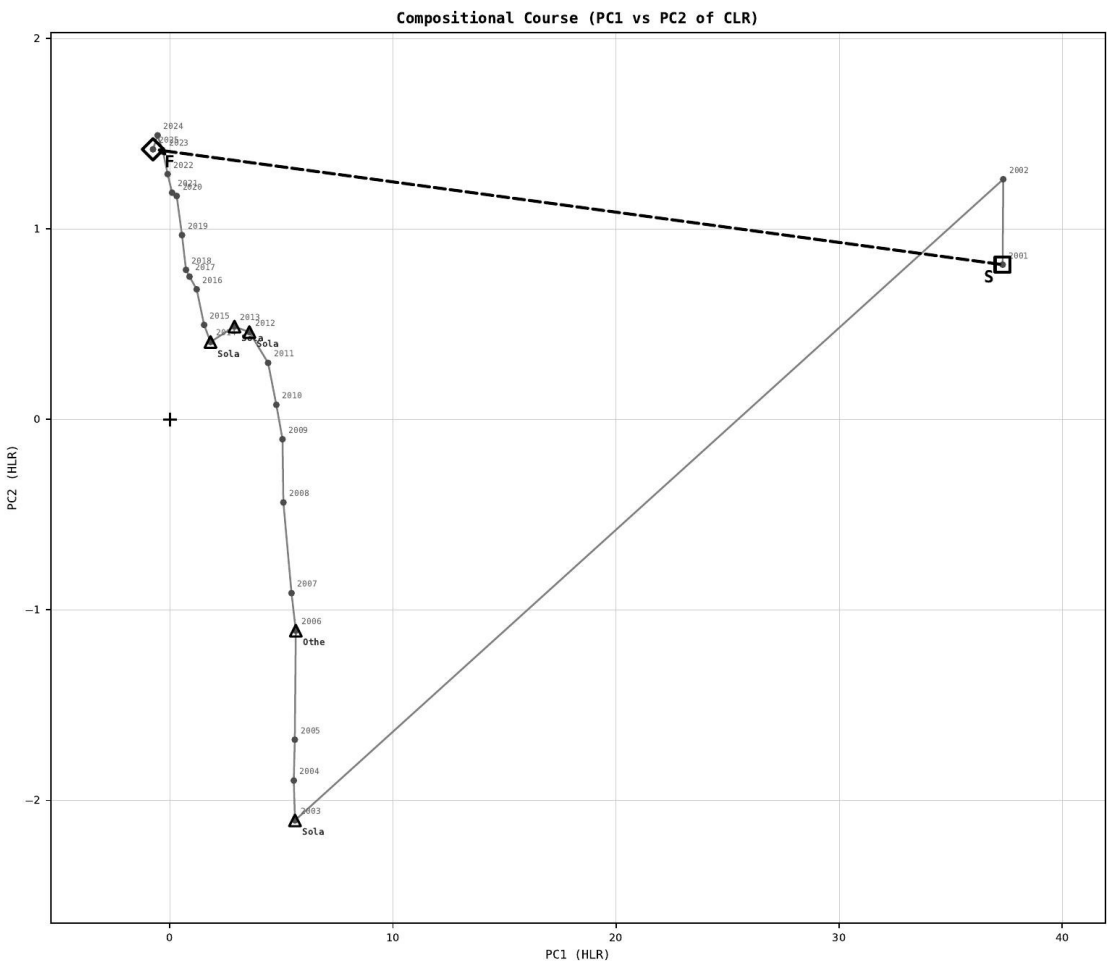
ABSENT

nuclear-stable

United States — the trajectory

United States: the full trajectory — a large grid; no deceptive drift at annual grain.

SYSTEM COURSE PLOT | ember_USA_United_States_generation_TWh.csv | D=9 N=25 | Years 2001–2025



COURSE METRICS	
Scale Provenance: Y_j → x_j=Y_j/T → h_j=CLR → PC projection Unit: HLR (Higgins Log-Ratio)	
Net distance	= 38.0885 HLR
Path length	= 41.8379 HLR
Directness	= 0.9104
DR start	= 42.090 HLR
ratio	= 1903969992697630976.0x
DR final	= 4.633 HLR
ratio	= 102.8x
E_metric start	= 1411.9935
E_metric final	= 20.9704
PC1 loadings:	
Solar	-0.938
Other Fo	+0.159
Coal	+0.146
PC2 loadings:	
Wind	+0.831
Other Fo	-0.385
Coal	-0.324
Top displacement events:	
2003 Solar	d=31.925
2014 Solar	d=1.080
2012 Solar	d=0.882
2006 Other Fo	d=0.708
2013 Solar	d=0.691
The instrument reads. The expert decides. The loop stays open.	

USA

deceptive drift

ABSENT

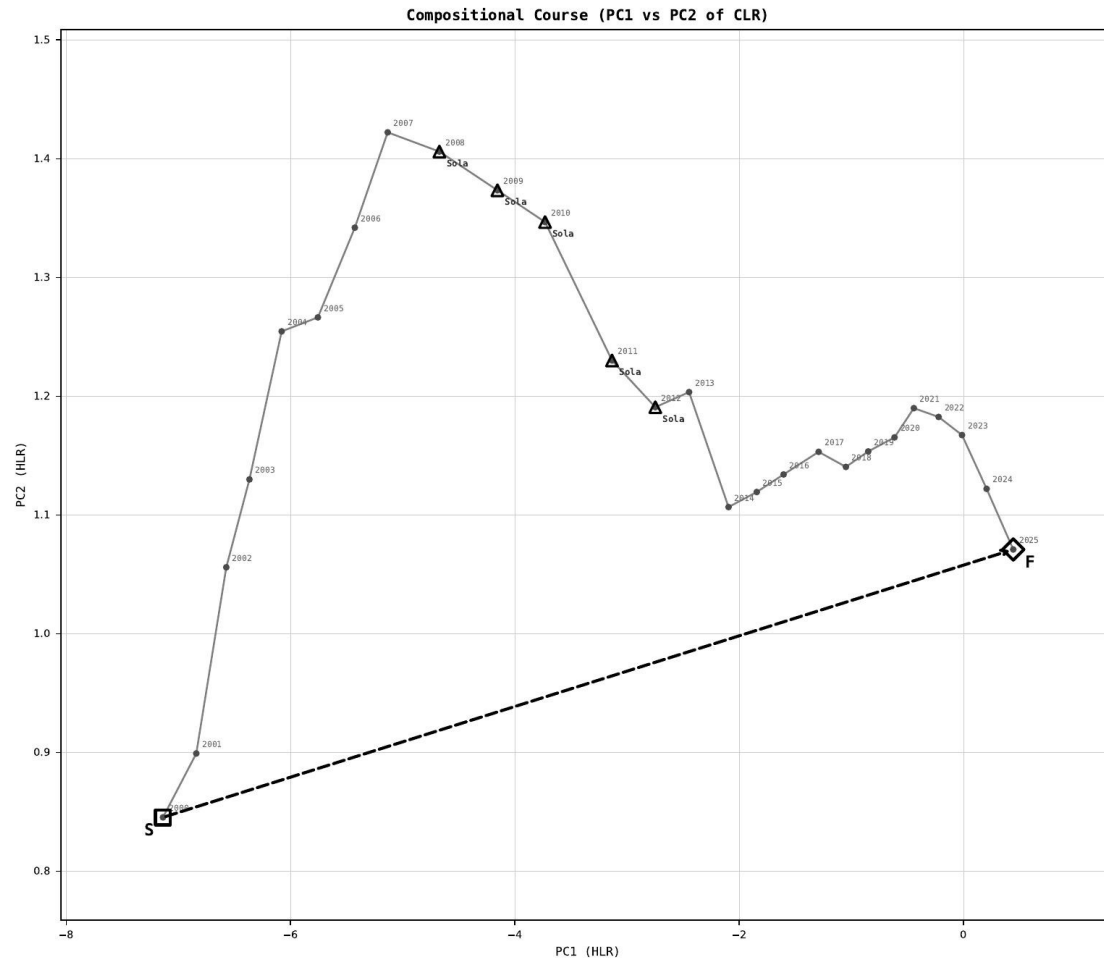
large grid

annual grain

World — the trajectory

World: the aggregate trajectory — large-N smoothing hides the deceptive drift in its constituents.

SYSTEM COURSE PLOT | ember_WLD_World_generation_TWh.csv | D=9 N=26 | Years 2000–2025



COURSE METRICS	
Scale Provenance: $Y_j \rightarrow x_j = Y_j/T \rightarrow h_j = CLR$ $\rightarrow$ PC projection Unit: HLR (Higgins Log-Ratio)	
Net distance	= 7.5841 HLR
Path length	= 7.9588 HLR
Directness	= 0.9529
DR start	= 8.647 HLR
ratio	= 5694.9x
DR final	= 4.764 HLR
ratio	= 117.2x
E_metric start	= 66.7408
E_metric final	= 16.7040
PC1 loadings:	
Solar	+0.816
Wind	+0.324
Other Fo	-0.296
PC2 loadings:	
Wind	+0.827
Solar	-0.443
Other Fo	-0.193
Top displacement events:	
2011 Solar	d=0.629
2009 Solar	d=0.520
2008 Solar	d=0.464
2010 Solar	d=0.430
2012 Solar	d=0.401

The instrument reads.
The expert decides.
The loop stays open.

WLD

deceptive drift

ABSENT

large-N

smoothing

What the stack answers

Recap of the five readings — then the live close: the CN-TT Output PDF flashes the raw plates, then the projector takes over.

WHAT

carriers are big.

size view

WHO

is in largest motion.

Helmsman

WHEN

the largest-motion carrier changes.

Helmsman trajectory

HOW MUCH

structural work each carrier did.

Power Share

WHY

a small carrier mattered.

Activation Coefficient

The stack does not replace interpretation. It gives interpretation a reproducible object.

Live close → 30 sec CodaWork2026_CN-TT_Output_2026-05-28.pdf (Stage 1 plates, movie-like) + 30 sec
codawork2026_projector.html

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